

Guided Mastery — Product Case Study

Designing a human-in-the-loop learning system (built with an AI partner)

The problem

A capable, time-strapped adult learner kept landing at “approaching competency” on a high-stakes, self-paced graduate exam despite real effort — overwhelmed, under-confident, and pouring hours into the wrong places. The failure wasn't missing content; it was a system that rewarded *recognizing* reused practice questions instead of building understanding that survives a reworded exam. A classic product trap: high-effort users still failing because the product optimizes the wrong metric.

The insight

Recognition ≠ mastery, and passing a practice test ≠ readiness. Real success has to be measured by **confidence-calibrated correctness**, **transfer** to novel items, and **retention** — not raw scores.

What I built (with an AI build partner)

- **Diagnose from real evidence** — parse results, response times, and stated confidence; classify every miss as a wrong mental model, a look-alike confusion, or a slip.
- **Prioritize by leverage** — impact × the learner's gap × tractability, not by how “important” a topic sounds.
- **Scaffold** — immediate-feedback practice with the governing principle in every explanation, plus mnemonics in the learner's own words.
- **Bake in evals** — verify every figure in code; use transfer tests (reworded, novel items) as the eval that exposes recognition masquerading as mastery.
- **Build confidence honestly** — reframe real assets, mark the smallest first step, never sell false hope.

The AI-product decisions (the Principal-level part)

- **Division of labor**: the system builds the scaffolds; the human authors the hooks and walks the path — which also keeps authorship and IP with the human.
- **Evaluation-first**: transfer tests and programmatic verification are the guardrail against a fluent-but-wrong model.
- **Metrics most teams miss**: sure-and-right, transfer, and decay — instrumented deliberately.
- **Guardrails**: academic integrity, honest-not-hype, and clear content ownership.

Results

On reworded, novel items, misses fell from 23 to 14 across attempts; the single most persistent error was converted and held up on fresh phrasings; entire weak areas recovered. The same framework then transferred cleanly to an entirely different domain — a person's career and life goal — proving it's a framework, not a one-off. It's now productized into installable skills and a self-guided atlas.

Why it matters for this role

A compact demonstration of Principal AI PM work: find the real problem behind the literal ask, prioritize by leverage, design AI systems with evals and guardrails, instrument for the metrics that actually signal value, keep the human in control, and ship something reusable.

One-line version: *I turned “I keep failing this exam” into a reusable, eval-backed learning system — by solving the felt problem (transfer, not recognition), prioritizing by leverage, and keeping a human in authorship.*